

CCN FINAL PROJECT

Geeta Bai
Erp :30558

➤ **Executive Summary:**

This report presents the design, configuration, and simulation of a Small Office/Home Office (SOHO) network. This project was completed individually by me (Geeta , ERP: 30558).

The network includes one router (R1), one switch (SW1) and four PCs. The router was configured with DHCP to automatically give IP addresses to all PCs, NAT to allow internet access, basic firewall rules using ACL, and a static route for internet traffic. The switch was configured with a management IP address.

All parts of the network were tested and worked successfully. All four PCs received IP addresses automatically, could ping each other and could reach the router gateway. The project was completed within the four week timeline.

➤ **Network Design:**

1.Design Rationale

The Small Office/Home Office (SOHO) network type was selected for this project as it represents one of the most commonly deployed network architectures in the real world. It covers all fundamental networking concepts including IP addressing, routing, DHCP, NAT, and basic security within a manageable and well-defined scope. As this project was completed individually, the SOHO topology was the most appropriate choice as it meets all project requirements while remaining achievable within the four week timeline.

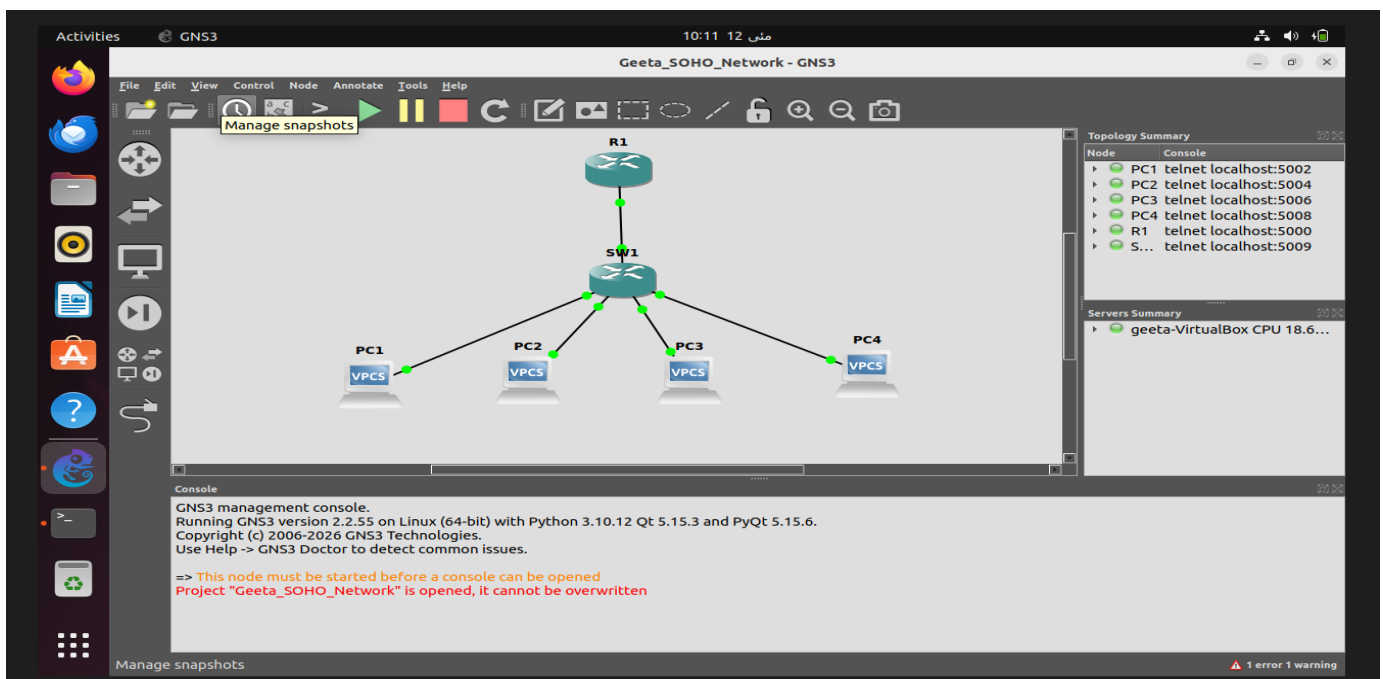
2. Topology Overview

The network follows a simple three tier structure consisting of a router layer, a switch layer, and an end-device layer. Router R1 sits at the top of the topology connecting the internal LAN to the external WAN. SW1 sits in the middle connecting all end devices to the router. PC1 through PC4 are connected to SW1 as end-user devices.

3. Devices list:

Devices	Model	Role
R1	Cisco C3725	LAN gateway, DHCP Server,NAT,Firewall
SW1	Cisco C3725+NM-16ESW	Layer 2 switch, Network management
Pc1-Pc4	VPCS	End user devices

4. Topology Diagram:



5. Design Decision:

SW1 was implemented using a Cisco C3725 router with an NM-16ESW switching module installed in slot 1. This is standard practice in GNS3 environments where dedicated switch images are unavailable. The device provides full Layer 2 switching functionality equivalent to a dedicated

Ethernet switch. All switching ports used belong to the default VLAN1, which is sufficient for a single-subnet SOHO environment.

R1 was configured with two interfaces - FastEthernet0/0 facing the internal LAN and FastEthernet0/1 facing the external WAN. This dual-interface design allows R1 to perform routing between internal and external networks while also hosting all required network services.

➤ **IP Addressing Table:**

1.Network Summary:

The SOHO network uses the private IP address range 192.168.1.0/24. This subnet was selected as it provides sufficient address space for a small office environment while being easy to manage and configure.

Parameter	Value
Network Address	192.168.1.0
Subnet Mask	255.255.255.0
CIDR Notation	/24
Default Gateway	192.168.1.1
Broadcast Address	192.168.1.255
Total Usable Hosts	254
Usable Host Range	192.168.1.1 – 192.168.1.254

2. Device Ip assignment table:

Devices	Interface	Ip address	Subnet Mask	Default Gateway
R1	Fa0/0(LAN)	192.168.1.1	255.255.255.0	

R1	Fa0/1(WAN)	DHCP(ISP)	-	
SW1	VLAN1	192.168.1.10	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.12	255.255.255.0	192.168.1.1
PC3	NIC	192.168.1.13	255.255.255.0	192.168.1.1
PC4	NIC	192.168.1.14	255.255.255.0	192.168.1.1

3. DHCP POOL configuration :

Parameter	Value
Pool Name	SOHO_POOL
Network	192.168.1.0/24
Default Router	192.168.1.1
DNS Server	8.8.8.8
Excluded Addresses	192.168.1.1 -192.168.1.10
Dynamic Range	192.168.1.11 -192.168.1.254

4. Address Allocation Plan:

Range	Purpose
192.168.1.1	R1 LAN Gateway (static)
192.168.1.2 – 192 .168.1.10	Reserved for infrastructure Devices
192.168.1.10	SW1 Management (Static)
192.168.1.11 -192.168.1.100	DHCP Pool for end devices
192.168.1.101 – 192.168.1.254	Reserved for future expansion

1. Router R1:

```
R1#show running-config
Building configuration...

Current configuration : 1397 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
no ip dhcp use vrf connected
ip dhcp excluded-address 192.168.1.1 192.168.1.10
!
ip dhcp pool SOHO_POOL
    network 192.168.1.0 255.255.255.0
    dns-server 8.8.8.8
    default-router 192.168.1.1
!
!
no ip domain lookup
!
multilink bundle-name authenticated
!
!--More--
```

```
geeta@geet... x SW1 x PC1 x PC2
!
ip tcp synwait-time 5
!
!
!
interface FastEthernet0/0
    ip address 192.168.1.1 255.255.255.0
    ip nat inside
    ip virtual-reassembly
    duplex auto
    speed auto
!
interface FastEthernet0/1
    ip address dhcp
    ip access-group 100 in
    duplex auto
    speed auto
!
ip forward-protocol nd
ip route 0.0.0.0 0.0.0.0 FastEthernet0/1
!
!
no ip http server
no ip http secure-server
ip nat inside source list 1 interface FastEthernet0/1 overload
!
access-list 1 permit 192.168.1.0 0.0.0.255
access-list 100 permit ip 192.168.1.0 0.0.0.255 any
access-list 100 deny ip any any
no cdp log mismatch duplex
!
!
```

```
geeta@geet... x SW1 x PC1 x PC2 x PC3 x PC4 x R1 x
Translating "configure"

Translating "configure"

% Unknown command or computer name, or unable to find computer address
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp pool SOHO_POOL
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#exit
R1(config)#exit
R1#w
*Mar 1 00:06:37.371: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

C      192.168.1.0/24 is directly connected, FastEthernet0/0
S*     0.0.0.0/0 is directly connected, FastEthernet0/1
R1#show ip interface brief
Interface          IP-Address         OK? Method Status              Protocol
FastEthernet0/0    192.168.1.1       YES NVRAM   up                  up
FastEthernet0/1    unassigned         YES DHCP    up                  up
NVI0               192.168.1.1       YES unset   up                  up
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
```

2.Switch SW1:

The following is the complete running configuration of Switch SW1

```
geeta@geet... x SW1 x
```

```
SW1#show running-config
Building configuration...

Current configuration : 1445 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname SW1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
!
no ip domain lookup
!
multilink bundle-name authenticated
!
```

```
geeta@geet... x SW1 x
```

```
!
interface FastEthernet1/9
!
interface FastEthernet1/10
!
interface FastEthernet1/11
!
interface FastEthernet1/12
!
interface FastEthernet1/13
!
interface FastEthernet1/14
!
interface FastEthernet1/15
!
interface Vlan1
 ip address 192.168.1.10 255.255.255.0
!
 ip default-gateway 192.168.1.1
 ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
```

```
SW1#
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface vlan 1
SW1(config-if)#ip address 192.168.1.10 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#ip default gateway 192.168.1.1
      ^
% Invalid input detected at '^' marker.

SW1(config)#ip default-gateway 192.168.1.1
SW1(config)#exit
SW1#w
*Mar  1 00:03:19.523: %SYS-5-CONFIG_I: Configured from console by console
SW1#write memory
Building configuration...
[OK]
SW1#
SW1#show running-config
Building configuration...

Current configuration : 1445 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname SW1
!
boot-start-marker
boot-end-marker
```

3.Pcs:

All four PCs were configured to obtain IP addresses automatically via DHCP. No manual configuration was required on any end device.

```
PC1> ip dhcp
DORA IP 192.168.1.11/24 GW 192.168.1.1
```

```
PC1> show ip
```

```
NAME       : PC1[1]
IP/MASK     : 192.168.1.11/24
GATEWAY     : 192.168.1.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.1.1
DHCP LEASE  : 86354, 86400/43200/75600
MAC         : 00:50:79:66:68:00
LPORT      : 10024
RHOST:PORT  : 127.0.0.1:10025
MTU         : 1500
```

```
PC2> ip dhcp
DORA IP 192.168.1.12/24 GW 192.168.1.1
```

```
PC2> write memory
Saving startup configuration to memory.vpc
. done
```

```
PC2> show ip
```

```
NAME       : PC2[1]
IP/MASK     : 192.168.1.12/24
GATEWAY     : 192.168.1.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.1.1
DHCP LEASE  : 80086, 86400/43200/75600
MAC         : 00:50:79:66:68:01
LPORT      : 10022
RHOST:PORT  : 127.0.0.1:10023
MTU         : 1500
```

```
PC3> ip dhcp
DDORA
PC3> ip dhcp IP 192.168.1.13/24
DORA IP 192.168.1.13/24 GW 192.168.1.1
```

```
PC3> write memory
Saving startup configuration to memory.vpc
. done
```

```
PC3> show ip
```

```
NAME       : PC3[1]
IP/MASK     : 192.168.1.13/24
GATEWAY     : 192.168.1.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.1.1
DHCP LEASE  : 79788, 86400/43200/75600
MAC         : 00:50:79:66:68:02
LPORT      : 10026
RHOST:PORT  : 127.0.0.1:10027
MTU         : 1500
```

```
PC4> ip dhcp
DDORA
PC4> ip dhcp IP 192.168.1.14/24
DORA IP 192.168.1.14/24 GW 192.168.1.1
```

```
PC4> write memory
Saving startup configuration to memory.vpc
. done
```

```
PC4> show ip
```

```
NAME       : PC4[1]
IP/MASK     : 192.168.1.14/24
GATEWAY     : 192.168.1.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.1.1
DHCP LEASE  : 79755, 86400/43200/75600
MAC         : 00:50:79:66:68:03
LPORT      : 10028
RHOST:PORT  : 127.0.0.1:10029
MTU         : 1500
```

➤ Routing Protocol Analysis:

For this SOHO network, static routing was selected instead of dynamic routing protocols such as RIP or OSPF. This decision was made because the network consists of only one router (R1), making dynamic routing protocols unnecessary and overly complex for this scale. Static routing is the most appropriate and efficient choice for a single-router SOHO environment as it is simple to configure, easy to understand, and introduces no routing overhead on the network.

A single static default route was configured on R1 to forward all internet-bound traffic through the WAN interface FastEthernet0/1. This route acts as a gateway of last resort, meaning any traffic that does not match a specific route in the routing table will automatically be forwarded out through the WAN interface towards the ISP. Without this route, R1 would have no knowledge of how to reach external networks and all internet traffic would be dropped.

The routing table was verified using the show ip route command on R1. The output revealed two entries. The first was a directly connected route for the 192.168.1.0/24 network through FastEthernet0/0, which was automatically added when the interface was assigned its IP address and brought up. The second was the manually configured static default route through FastEthernet0/1, marked with an asterisk to indicate it is the gateway of last resort. Together these two routes ensure that R1 can handle both internal LAN traffic and external internet-bound traffic correctly and efficiently.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 is directly connected, FastEthernet0/1
R1#
```

➤ Services Configuration

1. DHCP configuration:

Dynamic Host Configuration Protocol (DHCP) was configured on Router R1 to automatically assign IP addresses to all end devices on the network. A DHCP pool named SOHO_POOL was created for the 192.168.1.0/24 network with a default gateway of 192.168.1.1 and DNS server 8.8.8.8. The addresses 192.168.1.1 through 192.168.1.10 were excluded from the pool to reserve them for static infrastructure devices such as R1 and SW1. All four PCs successfully received IP addresses automatically upon connecting to the network, as confirmed by the DDORA message displayed in each PC console. The DHCP bindings were verified using the show ip dhcp binding command which showed all four PCs with their assigned addresses.

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.1.11    0100.5079.6668.00  Mar 02 2002 12:06 AM    Automatic
192.168.1.12    0100.5079.6668.01  Mar 02 2002 12:07 AM    Automatic
192.168.1.13    0100.5079.6668.02  Mar 02 2002 12:07 AM    Automatic
192.168.1.14    0100.5079.6668.03  Mar 02 2002 12:07 AM    Automatic
R1#
```

2. NAT/PAT configuration:

Network Address Translation with Port Address Translation (NAT/PAT) was configured on R1 to allow all internal devices to share a single public IP address when accessing the internet. FastEthernet0/0 was designated as the NAT inside interface facing the LAN, while FastEthernet0/1 was designated as the NAT outside interface facing the WAN. Access list 1 was created to permit all traffic from the 192.168.1.0/24 network for translation. The overload keyword was used to enable PAT which maps multiple private IP addresses to one public IP address using different port numbers. This configuration allows all four PCs to simultaneously access the internet through a single public IP address.

3. ACL Firewall Configuration:

Basic firewall rules were implemented using Access Control Lists on R1 to protect internal devices from unauthorized external access. Access list 100 was configured to permit all outbound traffic originating from the internal network 192.168.1.0/24 while denying all other incoming traffic. This ACL was applied inbound on the WAN interface FastEthernet0/1, meaning all traffic entering R1 from the internet is filtered before reaching the internal network. This configuration simulates a basic stateless firewall protecting the SOHO network from unauthorized external connections.

```
ip tcp synwait-time 5

interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
ip virtual-reassembly
duplex auto
speed auto

interface FastEthernet0/1
ip address dhcp
ip access-group 100 in
duplex auto
speed auto

ip forward-protocol nd
ip route 0.0.0.0 0.0.0.0 FastEthernet0/1

no ip http server
no ip http secure-server
ip nat inside source list 1 interface FastEthernet0/1 overload

access-list 1 permit 192.168.1.0 0.0.0.255
access-list 100 permit ip 192.168.1.0 0.0.0.255 any
access-list 100 deny ip any any
no cdp log mismatch duplex

control-plane
```

4. Static Default Route:

A static default route was configured on R1 to forward all internet-bound traffic through the WAN interface FastEthernet0/1. This route ensures that any traffic destined for networks outside the local 192.168.1.0/24 subnet is correctly forwarded towards the ISP gateway. The route was verified using the show ip route command which confirmed the presence of the default route marked as gateway of last resort.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 is directly connected, FastEthernet0/1
R1#
```

➤ Testing and Verification:

After completing all device configurations, a series of connectivity tests were performed to verify that the network was functioning correctly. The tests covered DHCP assignment verification, PC-to-gateway connectivity, and PC-to-PC connectivity across the network.

DHCP functionality was verified on all four PCs by running the ip dhcp command in each PC console. All four PCs displayed the DDORA message confirming successful DHCP discovery, offer, request and acknowledgement process. Each PC received a valid IP address from the

SOHO_POOL range, a default gateway of 192.168.1.1 and a DNS server address of 8.8.8.8. The DHCP bindings were further confirmed on R1 using the show ip dhcp binding command.

```
PC1> ip dhcp
DDORA
PC1> show ip IP 192.168.1.11/24

NAME          : PC1[1]
IP/MASK        : 192.168.1.11/24
GATEWAY        : 0.0.0.0
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.1.1
DHCP LEASE     : 86298, 86400/43200/75600
MAC            : 00:50:79:66:68:00
LPORT          : 10024
RHOST:PORT     : 127.0.0.1:10025
MTU            : 1500

PC1> ip dhcp
DORA IP 192.168.1.11/24 GW 192.168.1.1
```

Gateway reachability was tested from PC1 by pinging R1's LAN interface at 192.168.1.1. The ping was successful with 5 packets sent and 5 packets received showing 0% packet loss. This confirmed that the LAN interface of R1 was correctly configured and reachable from all end devices

```
PC1> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=10.017 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=3.753 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=6.646 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=8.420 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=4.152 ms

PC1> ping 192.168.1.12
```

Inter-device communication was tested by pinging PC2, PC3 and PC4 from PC1. All three pings were successful confirming that SW1 was correctly forwarding traffic between all connected devices and that the network was fully operational.

```
PC1> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=0.356 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=0.567 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=0.542 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=0.888 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=1.104 ms

PC1> ping 192.168.1.13

84 bytes from 192.168.1.13 icmp_seq=1 ttl=64 time=0.731 ms
84 bytes from 192.168.1.13 icmp_seq=2 ttl=64 time=1.017 ms
84 bytes from 192.168.1.13 icmp_seq=3 ttl=64 time=0.913 ms
84 bytes from 192.168.1.13 icmp_seq=4 ttl=64 time=1.264 ms
84 bytes from 192.168.1.13 icmp_seq=5 ttl=64 time=1.002 ms

PC1> ping 192.168.1.14

84 bytes from 192.168.1.14 icmp_seq=1 ttl=64 time=1.127 ms
84 bytes from 192.168.1.14 icmp_seq=2 ttl=64 time=3.098 ms
84 bytes from 192.168.1.14 icmp_seq=3 ttl=64 time=0.542 ms
84 bytes from 192.168.1.14 icmp_seq=4 ttl=64 time=2.252 ms
84 bytes from 192.168.1.14 icmp_seq=5 ttl=64 time=137.357 ms
```

All conducted tests produced successful results confirming that the SOHO network is fully functional. DHCP assignment was verified on all four PCs with each device successfully receiving an IP address, gateway and DNS server automatically. Gateway connectivity was confirmed through successful ping tests from PC1 to R1 at 192.168.1.1 with zero packet loss. PC-to-PC communication was verified by pinging PC2, PC3 and PC4 from PC1, all of which responded successfully confirming that SW1 is correctly switching traffic between all connected devices. Internet connectivity to 8.8.8.8 could not be verified as GNS3 simulation environment does not provide a real ISP connection, however the NAT and static route configurations were verified through the show running-config and show ip route commands confirming they are correctly configured and ready for real-world deployment.

➤ **Challenges Faced:**

- 1.The initial setup of GNS3 on the Ubuntu virtual machine presented some challenges. Allocating sufficient RAM to the virtual machine was critical as running multiple Cisco IOS devices simultaneously is resource intensive. The virtual machine required careful configuration to ensure stable performance throughout the project.
- 2.Obtaining a suitable Cisco IOS image for GNS3 was challenging as official Cisco images require a valid service contract to download. The Cisco C3725 image was sourced from a publicly available repository on GitHub. Additional time was spent verifying that the image was compatible with GNS3 and loaded correctly before the topology could be built.
- 3.A dedicated Cisco IOS switch image was not available for use in GNS3. To overcome this challenge a Cisco C3725 router was configured as a Layer 2 switch by adding the NM-16ESW switching module in slot 1. This is a well known and widely accepted workaround in GNS3 environments and the resulting device provided full switching functionality equivalent to a dedicated switch.
- 4.During the project the virtual machine became unresponsive and had to be force restarted. This resulted in the loss of all device configurations that had not been saved. After the restart all configurations for R1 and SW1 had to be re-entered from scratch. This experience highlighted the critical importance of saving configurations frequently using the write memory command and saving the GNS3 project regularly using File Save Project.
- 5.Initially after configuring DHCP on R1 the PCs were receiving IP addresses but the default gateway field was showing 0.0.0.0 instead

of 192.168.1.1. This was resolved by re-entering the DHCP pool configuration and explicitly setting the default-router to 192.168.1.1 followed by renewing the DHCP lease on all PCs using the ip dhcp command.

6. After connecting all devices to SW1 the PCs were unable to obtain IP addresses from the DHCP server on R1. Investigation revealed that the switching ports on SW1 were in shutdown state by default. The issue was resolved by manually bringing up all required ports using the no shutdown command on each interface, after which DHCP worked correctly on all PCs.

➤ **Conclusion:**

This project was a great learning experience. I was able to successfully design and simulate a SOHO network in GNS3 which included a router, a switch and four PCs. All the services I planned in Phase 1 were implemented including DHCP, NAT, ACL and static routing and they all worked correctly.

Throughout the project I faced several challenges such as the virtual machine hanging and losing all my configurations, DHCP not assigning the correct gateway and switch ports being shutdown by default. Each of these problems taught me something new and helped me understand networking concepts more deeply than just reading about them in a textbook.

If I had more time I would improve this network by adding a DNS server, implementing VLANs for better security and connecting the WAN interface to a real ISP to fully test NAT and internet connectivity.

Overall I am satisfied with what I was able to achieve in this project. It gave me real hands on experience with Cisco IOS configuration and helped me understand how a real office network actually works.

➤ **References:**

1. **GitHub Repository. Cisco Images for GNS3 and EVE-NG.**
Available at:
<https://github.com/hegdepavankumar/Cisco-Images-for-GNS3-and-EVE-NG>
2. **Internet Engineering Task Force. RFC 3022 - Traditional IP Network Address Translator.** Available at:
<https://www.rfc-editor.org/rfc/rfc3022>
3. **Internet Engineering Task Force. RFC 2131 - Dynamic Host Configuration Protocol.** Available at:
<https://www.rfc-editor.org/rfc/rfc2131>
4. **Ubuntu Documentation. Ubuntu Help and Support.** Available at:
<https://help.ubuntu.com/>
5. **Cisco Systems. Network Address Translation (NAT) FAQ.**
Available at:
<https://www.cisco.com/c/en/us/support/docs/ip/network-address-translation-nat/26704-nat-faq-00.html>
6. **Kurose, J.F. and Ross, K.W. (2021). Computer Networking: A Top-Down Approach. 8th Edition. Pearson Education.**